Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

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Scientific Context: Nonlinearity & Instability

The Challenge: Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

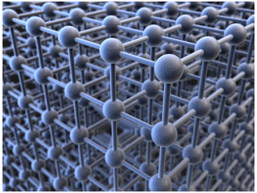
- **Phase Transitions:** Martensitic transformations.
- **Plasticity:** Dislocation-mediated flow and avalanches.
- **Fracture:** Brittle-to-ductile transitions and damage.

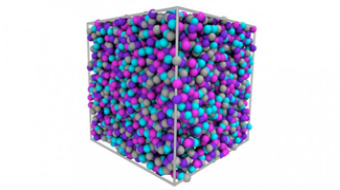
Unifying Perspective

Treating these diverse phenomena as *displacive instabilities* driven by the competition between elastic energy and lattice symmetry.

Non-linear phenomena

Crystals





Meta-materials